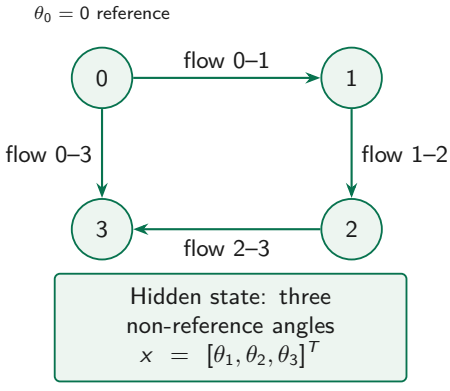
Sensors report numbers; the state is what we infer



Fresh four-bus DC fixture

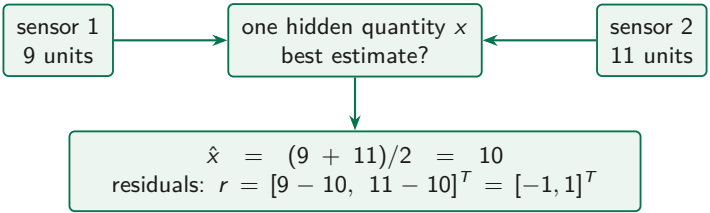
Eight measurements: four flows, three injections, one angle. Three unknown angles. Redundancy permits a consistency test.

Predict: Is a small residual the same as a trustworthy state?

Explanation: No. We will construct two different states with the same residual.
This identifiable DC example is separate from the three-phase distribution-feeder models.

Classroom_Labs L8–L9; standard_formulations.run_wls_demo. State estimation: Schweppe and Wildes, 1970.

Two equally precise sensors: find their common explanation



Predict: What mismatch remains after choosing the best common value?

Explanation: $J = (-1)^2 + 1^2 = 2$. A residual measures mismatch to the fitted model. Illustrative scalar example: $H = [1, 1]^T$, unit variance for each sensor. This is WLS with equal weights, not the four-bus run.

Original arithmetic teaching example; least-squares principle: Schweppe and Wildes, 1970.

An isolated error increases disagreement between sensors

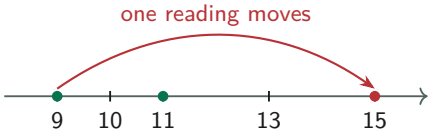
Before corruption

$z = [9, 11]^T$
$\hat{x} = 10$
$r = [-1, 1]^T$
$J = 2.$

Change only sensor 1

$z' = [15, 11]^T$
Which common estimate fits best?

$$\hat{x}' = 13, \quad r' = [2, -2]^T, \quad J' = 8.$$



A calibrated alarm signals model inconsistency; it does not identify an attacker.

Original scalar illustration, equal unit variances. Statistical alarm assumptions are introduced on C06.

Coordinated errors can preserve the same disagreement

	Measurements	Fitted state	Residual
Original	$[9, 11]^T$	10	$[-1, 1]^T$
Both shifted by $+3$	$[12, 14]^T$	13	$[-1, 1]^T$

Same residual statistic: $J' = J = 2$.

Different inferred state: $\hat{x}' = \hat{x} + 3$.

Consistency is not authenticity

Both readings fit a shifted state. Consistency alone cannot recover the missing truth.

Predict: Does LLM voting add an independent measurement?

Explanation: No. New physical evidence must come from outside the corrupted readings.

Original scalar adaptation of the false-data-injection principle in Liu, Ning and Reiter, CCS 2009.

Weighted least squares fits the measurement model

$$z = Hx + e, \qquad \hat{x} = \arg\min_x (z - Hx)^T R^{-1} (z - Hx), \qquad R = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_m^2).$$

$z \in \mathbb{R}^m$	sensor measurements
$x \in \mathbb{R}^n$	unknown state
$H \in \mathbb{R}^{m \times n}$	linear measurement map
e	measurement noise
σ_j	channel j noise scale
R	noise covariance

Our fresh network run

$m = 8$, $n = 3$, $\mathrm{rank}(H) = 3$.
Flow $\sigma = 0.004$ p.u.; injection 0.006 p.u.; angle 0.0005 rad.

Higher precision means larger weight. Each discrepancy is normalized by its own noise scale.

Classroom_Labs L8; standard_formulations.weighted_least_squares. WLS: Schweppe and Wildes, 1970.

A residual alarm is a statistical test under stated assumptions

$$r = z - H\hat{x}, \qquad J = r^T R^{-1} r, \qquad J > \chi^2_{m-n,\,1-\alpha} \implies \text{alarm.}$$

Classroom calibration

$m - n = 8 - 3 = 5$ residual degrees of freedom. $\alpha = 0.01$; threshold $\chi^2_{5,0.99} = 15.0863$.

Interpretation requires assumptions

Correct linear model; full column rank; stated independent Gaussian noise and covariance. α is the nominal false-alarm probability.

Predict: Does "no alarm" mean "safe to control"?

Explanation: No. It means this test found no excessive model inconsistency. Provenance and operating limits remain separate checks.

Residual test implementation: standard_formulations.weighted_least_squares. State estimation foundations: Schweppe and Wildes, 1970.

The cancellation is visible in three lines

$$\text{Coordinated measurement change:} \quad z' = z + Hc,$$

$$\text{New WLS estimate:} \quad \hat{x}' = \hat{x} + c,$$

$$\text{New residual:} \quad r' = z + Hc - H(\hat{x} + c) = z - H\hat{x} = r.$$

$c \in \mathbb{R}^n$: chosen state shift; $a = Hc$: coordinated change to sensor readings. Therefore $J' = J$ with unchanged covariance.

The assumptions are part of the result

Full-rank linear WLS; correct knowledge of H; ability to alter the required unprotected channels. A protected channel can restrict feasible shifts.

Predict: Which term disappears when we subtract the predicted measurements?
The added Hc cancels exactly.

False-data-injection construction: Liu, Ning and Reiter, CCS 2009. Defensive four-bus exercise in L8.

Hands-on: sweep isolated and coordinated corruption

Notebook L8 Seven minutes: run the full cell; compare residual and state-error plots.

```
ATTACK_SCALES = (0., .25, .5, 1., 1.5)
started = time.perf_counter()
cyber = lab.cyber_sweep(ATTACK_SCALES)
timings['L8 fresh WLS attack sweep'] = time.perf_counter() - started
display(cyber['table'])
```

Predict first

For a growing isolated bias, what happens to J? For a growing Hc shift, what happens to J and the state error?

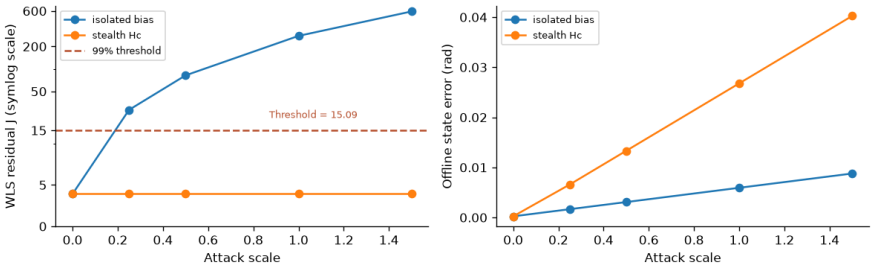
Read two different quantities

The residual is observable from measurements and the model. State error uses simulated truth and is an offline evaluation metric.

The scale multiplies a synthetic perturbation. It is not an estimate of attack likelihood or real-world severity.

Exact excerpt: Classroom_Labs L8 lines 1–5. Fixed synthetic seeds and attack directions in src/classroom_labs.py and standard_formulations.py.

Same residual does not imply the same state



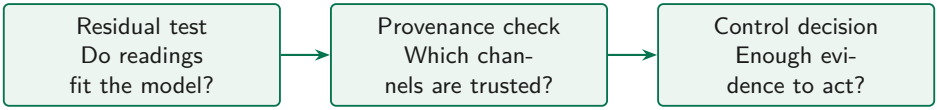
At scale 1: isolated-corruption $J \approx 277.50$; coordinated-shift $J \approx 3.92$. Threshold 15.09.

Keep the metrics separate

Left: consistency evidence available to the estimator. Right: state error against simulated truth, used only to evaluate the experiment.

Original computed figure: Classroom_Labs L8 / classroom_cyber.png. Coordinated $J \approx 3.92022$ remains constant across the sampled scales.

Awareness must distinguish consistency from trustworthy provenance



A legitimate output is HOLD

If measurements are suspect and no trustworthy replacement is available, request evidence instead of manufacturing a confident state estimate.

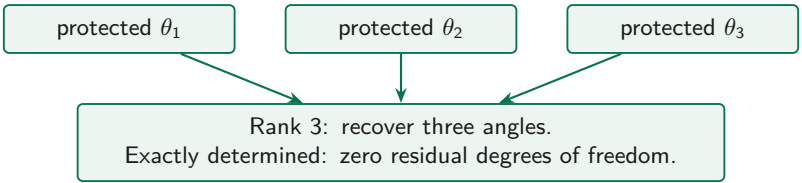
Predict: Can an LLM vote resolve missing observability or missing trusted channels?

Explanation: No. Voting over the same evidence does not add an independent physical constraint.

Information limitation illustrated by Liu et al., 2009; explicit protected-channel assumption in Classroom_Labs L9.

Recovery assumes three independent protected angle readings

$$z_T = H_T x + e_T, \qquad H_T = I_3, \qquad \sigma_T = 0.00005 \text{ rad per channel.}$$



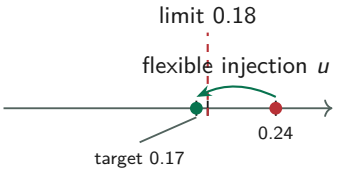
Subscript T: trusted set; I_3: identity matrix; e_T: assumed measurement noise.

An assumption, not an implemented security protocol

Authentication and physical protection are external requirements. This three-channel fit provides no redundant chi-square detection test.

Exact classroom recovery model: standard_formulations.run_wls_demo; Classroom_Labs L9. Protected measurements are assumed trustworthy.

Corrective control: move a stressed flow back inside its limit



Noiseless arithmetic illustration

For branch 1–2, suppose estimated flow is 0.24 p.u. and sensitivity is $-4/7$.
A 0.01 p.u. margin gives target 0.17.

$u \geq (0.24 - 0.17)/(4/7) = 0.1225$ p.u.

u: injection at node 2, balanced by the reference source. The actual noisy protected-state run gives $u \approx 0.12169$ p.u.; the illustration above is not that exact solve.

Sensitivity $FB^{-1}[0, 1, 0]^T$ from standard_formulations._redispatch. Original noiseless arithmetic illustration; all quantities here are per unit.

Planning uses a constrained quadratic control problem

$$\min_{0\le u\le 0.5} \ \tfrac{1}{2}u^2 \quad \text{s.t.} \quad |F\hat{x}_T + b_u u| \le \ell - \delta\mathbf{1}, \qquad b_u = FB^{-1}e_2.$$

F: angle-to-flow map; B: non-reference nodal injection matrix; $e_2 = [0,1,0]^T$; $\hat{x}_T$: trusted estimate; b_u: flow sensitivity.
$\ell = [0.35, 0.18, 0.35, 0.35]^T$ p.u.: branch limits; $\delta = 0.01$ p.u.: planning margin; $\mathbf{1}$: vector of ones. Absolute values and inequalities are componentwise.

Notebook L9 Run both evidence conditions with the same numerical control method.

```
without_trust = lab.trusted_response(cyber, protected_available=False)
with_trust = lab.trusted_response(cyber, protected_available=True)
```

Method and acceptance

SLSQP solves this scalar convex QP. Independent post-solve bounds and predicted-flow checks can still return HOLD.

Exact excerpt: Classroom_Labs L9 lines 1–2; constrained model and checks in standard_formulations._redispatch. DC illustration, not AC feeder resilience.

Hands-on: evidence availability changes the permitted response

Notebook L9 Five minutes: run the full cell; identify the acceptance assumption.

Condition	Response	Flexible injection
No trusted replacement channels	HOLD	none proposed
Three protected angles assumed	PROPOSE	0.12169 p.u.

Two checks, with different information

Predicted branch 1–2 flow: 0.17000 p.u. (operational model check).
Actual simulated flow: $0.17046 < 0.18$ p.u. (offline truth audit).

Predict: Why is the offline value not an input to the operator's verification rule?

Explanation: The real system does not reveal its exact state just because our simulator knows it.

Fresh L9 output with fixed synthetic seeds; standard_formulations.run_wls_demo. The zero-violation offline audit applies to this modeled case only.

Every stage has a specific numerical or evidentiary job

Stage	Specific method	Inspectable output
Awareness	WLS, residual and trust checks	Alarm, rank, provenance status
Planning	Constrained DC redispatch QP	Injection and predicted flow margins
Simulation	DC flow calculation	Response to the candidate injection
Verification	Provenance and limit gates	PROPOSE or HOLD, with failed condition
LLM Optimizer	Allowed refinement selection	New tool request and returned rationale

Use the method suited to the decision

WLS estimates the state; the QP selects the correction. L10 replays the archived five-stage language trace.

Classroom_Labs L8–L10; archived cyber_stealth / dc run in proposal_complete_workflow.json. Fresh numerical fixtures and archived LLM calls are separately labeled.

Actual Llama response: request a new checked candidate

Planning, 0.829 s

"Requesting new candidates based on the given tools and objective."

LLM Optimizer selects TRUSTED_QP, 1.195 s

"The current lowest-cost candidate E03 has a cost of 0.008671610999497877, but the permitted refinement TRUSTED_QP offers a new approach to minimize the objective. Selecting this operation will request a new simulation and verification."

Numerical result: $u = 0.12169$ p.u., cost 0.00740467.
Acceptance still depends on trusted-input and flow-limit gates.

Exact returned rationale fields: archived cyber_stealth / dc trace, proposal_complete_workflow.json. Model-call wall times; rounded only outside quotes.

A matched baseline avoids crediting the LLM for better information

Archived DC cyber response	Fixed supervisor	Llama supervisor
Flexible injection (p.u.)	0.12169	0.12169
Objective $\frac{1}{2}u^2$	0.00740467	0.00740467
Total elapsed time (s)	0.00381	5.203

What actually enabled recovery?

The trusted measurement assumption plus constrained numerical control. Llama added tool-selection language and a trace; it did not improve this recorded optimum.

Predict: Would comparing untrusted baseline data with trusted agent data be fair?

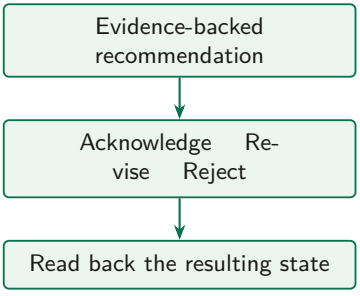
Explanation: No. That would mix the value of additional information with the value of orchestration.

ARCHIVED ACTUAL: fixed versus agentic cyber_stealth / dc, proposal_complete_workflow.json. Same objective and evidence assumptions; one recorded run, four Llama calls.

The control-room handoff is a decision packet, not a green badge

What the operator sees

Proposed action and objective.
Sensor trust and model scope.
Current revision and failed/passed checks.
Measured latency and available alternatives.



Predict: What did the three cases teach about choosing numerical methods?

Explanation: Continuous voltage control, discrete restoration and evidence-limited cyber response need different tools within one auditable workflow.

Case synthesis: Classroom_Labs L1–L10; operator readback exercise L12. The tutorial dashboard is an educational interface, not a deployed control system.